Calculations in WHERE

Lecture Notes · Game DB Junior

In this lesson

- We can do **calculations** inside a WHERE statement
- The **order of operations**: math first, then the comparison

1. A quick reminder: calculated columns

With SELECT, we can add **calculated columns**. For example, we calculate population divided by area and rename it to population_density:

```
SELECT name, population / area AS population_density
FROM cities;
```

When we run this, we see the population density of each city.

2. Calculations inside WHERE

We can do the **same kind of calculation** inside a WHERE statement too. We are **not** limited to checks on the existing columns in our table.

Example: find all the cities with a **population density greater than 6000**. We remove the semicolon at the end, add a WHERE clause, and write population / area > 6000:

```
SELECT name, population / area AS population_density
FROM cities
WHERE population / area > 6000;
```

3. Order of operations

There are two ways this could be understood:

Way	Meaning	Correct?
1	First do population / area, then check if the result is greater than 6000	**Yes**
2	Check if area itself is greater than 6000	No

Rule: do all the **math first**. So first we find the result of population / area, then we check if it is greater than 6000.

From the last query's output, Tokyo and Shanghai have a population density lower than 6000. So we expect only **Delhi** and **Sao Paulo**. When we run it, that is what we get.

Inside WHERE clauses we can add **as many** math operations, or even string operations, as we want. We can use all the operators we learned a few lessons ago.

4. Running the code (my practice)

Population density greater than 6000.

```sql
-- Show cities with a population density greater than 6000
SELECT
    name,
    population / area AS population_density
FROM
    cities
WHERE
    population / area > 6000;
```

Figure 1 — Calculation in SELECT and in WHERE

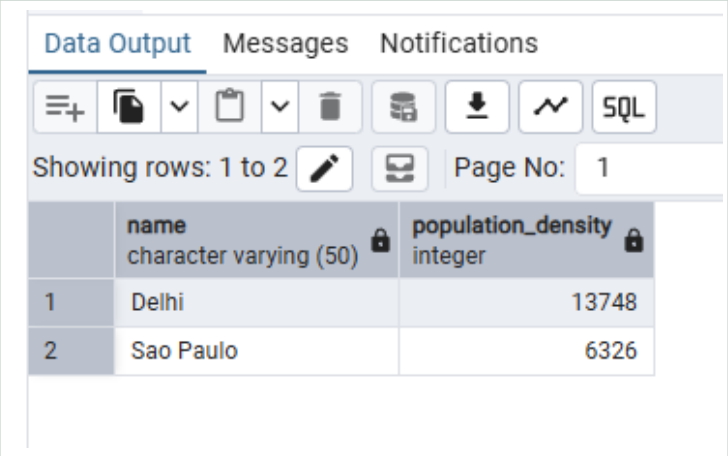


Figure 2 — Result: Delhi and Sao Paulo

5. Summary

Idea	Remember
Calculations in WHERE	We can check calculated values, not only existing columns
Order of operations	Math first, then the comparison
Operators	We can use math and string operators inside WHERE

Sources
Lecture transcript
My own practice screenshots (Figures 1–2)